

OPERA NUMERORUM

MODULE 8G WORMHOLE CERTIFICATE

120-Cell PCB Wormhole Certificate (ZoeM8G)
Battle Plan v1.6 -- David Fox -- June 2026

SHA-256 BINDING

SHA-256(m8g.out): 74a98aab034f604095504bc9f4fd1f0867c719c2da1b203e6069cc4db310672b
Source: certificates/m8g_wormhole_cert.py
depends_on: [M1, M5, M8B, M8C, M8D, M8F, M22]

THEOREM M8G (axiom_debt: [])

The 120-layer 10cm PCB resonator physically instantiates the H4-symmetric Morning Star transform. At $k_c = 3.183$, the cavity exhibits EM time contraction $dt_{internal} = dt_{external} / k_c$, falsifiable via the M8F 7-Layer Protocol.

1. Build Specification -- Version 2 PCB

Parameter	Value	Source	Tolerance
Frequency	299.314159265 MHz	M1 alpha_0	+/-10 Hz
Geometry	120-layer, 100mm dia	M8D V2	+/-0.1 mm
Layers	120 Cu, Rogers 4350B	M8D	0.1 mm each
Vias	720x 0.20 mm drill	120-cell faces	+/-10 um
Dihedral	116.565 deg	H4 exact	+/-0.01 deg
Q Factor	> 50,000	M8D	At 77 K

2. Certified Constants -- Inputs

Constant	Value	Source	Source SHA (prefix)
C_0	29.17 pF	M8B	a86219e55c181f2e...
C_cliff	166.98 pF	M8B	a86219e55c181f2e...
C_ratio	5.724374	M8B	a86219e55c181f2e...
Delta_DS^(4)	23.796910	M8B	a86219e55c181f2e...
c_bound	299541524 m/s	M8B	a86219e55c181f2e...
k_c	3.183	M22	5a5a345f6394438f...

3. Falsification Criteria -- M8F Protocol

1. **Cliff Test:** Sweep V_{drive} 0 to 5 V. Must observe C jump 29.17 to 166.98 pF at $k = 3.183 \pm 0.01$.

- 2. Pulse Test:** 0.5 m path. Must measure $t_{\text{cavity}} = 0.524 \pm 0.002$ ns vs $t_{\text{vacuum}} = 1.6678$ ns.
 $\Delta t_{\text{lead}} = 1.144$ ns early.
- 3. Null Result:** If $\Delta t \geq 1.667$ ns for all k then $v_g \leq c$. M8B dead. Report and archive.

4. Chain Consequences

If PASS	If FAIL
M8B, M22, M8F validated by physics. M23 BSD for J_0(143) gains experimental anchor via $c_{\text{bound}} = 299541524$. H4 symmetry 12/11 reduction physically implemented. Bost-Connes $C(S_4) = 11.4221$ is the mechanism for the capacitance cliff. GRH($X_0(143)$) chain has hardware root.	M8B falsified. depends_on chain breaks: M8D, M8F, M23 lose physical backing. BSD/GRH link to EM cavity severed. Axiom debt remains open. Report null result and archive.

5. Topology Correction

The 3D cross-section of the 120-cell cavity has **Poincare Homology Sphere (PHS)** topology: $\pi_1 = I^*$ (binary icosahedral group), $|I^*| = 120$, $H_1 = 0$. **NOT** lens space $L(5,1)$ [which has $H_1 = \mathbb{Z}/5\mathbb{Z}$]. Supervisor note corrected.

Property	PHS (correct)	$L(5,1)$ (wrong)
π_1	I^* order 120	$\mathbb{Z}/5\mathbb{Z}$
H_1	0 (I^* is perfect)	$\mathbb{Z}/5\mathbb{Z}$
Construction	Dodec + 36-deg twist	S^3 Dehn surgery
Euler chi (4D)	0 [PASS]	--
$ A_5 = 60$, $ I^* = 120$	A_5 simple $\Rightarrow I^*$ perfect	--

6. 10 Verified Identities

#	Identity	Status
1	$\alpha_0 \text{ MHz} = 299 + \pi/10$ [M1]	PASS
2	$f_{\text{res}} = 299.314159265\dots$ MHz [M1]	PASS
3	$C_{\text{ratio}} = C_{\text{cliff}} / C_0 = 5.724374$ [M8B]	PASS
4	$c_{\text{bound}} = 299541524$ [M8B]	PASS
5	$k_c = 3.183$ [M22]	PASS
6	$C(S_4) = 11.4221 > 2 \cdot \sqrt{13} = 7.2111$ [M5]	PASS
7	$t_{\text{vacuum}} = 1.6678$ ns [0.5m, c_{SI}]	PASS
8	$\Delta t_{\text{lead}} = 1.1440$ ns, err < 1% [M8F]	PASS
9	Euler chi = 0 [120-cell]	PASS

10	H_1(PHS) = 0, I* = 120 [topology]	PASS
----	-------------------------------------	------

ALL 10 ASSERTIONS PASS -- STATUS: SPEC_LOCKED

SHA Chain

Module	stdout SHA-256
M8G (this)	74a98aab034f604095504bc9f4fd1f0867c719c2da1b203e6069cc4db310672b
M8F	f7b049f9b18a182df5f41179ae0db85bc20c97a654e2d3d8c00775a64efc7418
M22	5a5a345f6394438f7a5134cf682d714fea6c89c73cfc22fcdc503bc90761e5ca
M8B	a86219e55c181f2e8ec7ee111188b8af0f2022a38a9a8dc18ba8335ebe828180
M8C	02fe604876c3253ec61ce0a8b382c7b01a089d1d217ab200fc9975464a645323
M8D	27d8e0c1e145ba7fb4a22c85067f3db78d92b490e592dcd255523afceec156db5

ASCII-only PDF: PASS -- no Unicode characters.
CERTIFIED -- Opera Numerorum -- M8G Wormhole Certificate